Ryan White

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A software and performance engineer with 12 years of experience in producing stable, performant software. Comfortable working across varied programming languages and platforms, and in deploying high throughput data platforms.

EXPERIENCE

One
Philadelphia, PA
www.one.app

One is a rising consumer financial services company backed by Ribbit and Walmart. One serves 4 million customers with the goal of financial progress for all.

Performance Engineer — Oct 2024 – Present

Sole full-time Performance engineer, tasked with optimizing performance of all company services, and scaling services to meet user growth. Working closely across SRE, developer, product, and external teams to improve experience for One customers. Worked with NodeJS applications, and reported bugs in functional typescript.

- Led organization and planning of team's short and long term goals.
- In weeks, performed load testing and configuration changes to assist in testing new Walmart integration API, used by 500k customers.
- Designed and created new Kubernetes autoscaling metrics based on requests per pod. This resulted in static response latencies for user profile services even as request volume rose 25x.

Nuix
Philadelphia, PA
www.nuix.com

Nuix is a leading data processing and analysis platform used worldwide by legal firms, governments, law enforcement agencies, and consulting firms.

Software Engineer — Feb 2023 – Oct 2024

Engineer on digital forensic and data visualization product, Nuix Investigate. Transitioned into software engineer role from previous role in performance engineering team. Developed new features and resolved bugs in Spring Boot, React, and data storage (Derby, Lucene, Elasticsearch) components.

- Implemented the backend API for a headline feature in the Fall 2023 Nuix Investigate release.
- Improved performance of communication preview feature by 8,000%, resolving several customer reports.
- Updated chat timeline view backend calls to allow for 100x more throughput, by reducing database activity and minimizing REST API response sizes.
- Consumed, aggregated, and exposed data stored in various locations like Apache Derby, Postgres, and Elasticsearch data stores.
- Led documentation updates, and created "single source of truth" for developer onboarding documentation.

- Wrote unit and integration tests in JUnit. Cleaned up broken, old, or redundant tests to reduce testing burden during development.

Senior Performance Engineer — Aug 2019 – Jan 2023
Performance Engineer — Nov 2015 – Aug 2019

Senior engineer on a key team responsible for the performance of all company products. Partnered with engineering, product, and sales teams for planning, solution optimization, and technical direction in the deployment of high throughput data platforms. Led team's short and long-term planning ranging from 2-week cycle planning to quarterly strategy planning, and helped translate business needs into engineering milestones.

- Designed, implemented, and owned company's first automated performance tests across all products. Developed reliable automated infrastructure on HyperV, and customized HP Loadrunner tool to work with automated, version-controlled tests. Improved release testing from a 2 week cycle involving distinct manual steps to an automated 4 hours.
- Developed and implemented a scaled Nuix REST solution capable of processing 1 TB of unstructured data per hour. This milestone is documented in a white paper available here: https://www.nuix.com/resources/scaling-nuix-rest-cluster-one-terabyte-hour.
- Supported seven-figure deployments for domestic and international clientsEngaged with clients in pre- and post-sales. Diagnosed customer issues and worked across international engineering teams to design and release patches.
- Worked directly with Nuix's founder and Chief Scientist on exploratory testing and optimization for core API shared across all products. Supported the shift of Nuix platform to scaled data storage with Elasticsearch, and then to networked Lucene indices networked through Derby Network Server.
- Initiated, led, and implemented cross-product process changes to "shift left" performance testing. This enabled performance testing of products during the development cycle instead of post-development, improving product quality and reliability.
- Improved response times for dozens of platform features by at least 10x, while reducing processor and memory usages.
- Managed and maintained $500k of local performance testing hardware, and effectively handled a quarterly AWS budget of $60k.

mcaConnect
Denver, CO
mcaconnect.com

mcaConnect is an award-winning consulting company and a Microsoft Gold Partner that implements custom solutions for clients working with Microsoft technologies.

Software Engineer — Jun 2014 – Nov 2015

Developer on the Delivery Services team, focusing on Microsoft Dynamics AX 2012. Responsible for development projects spanning multiple clients in manufacturing industries. Worked with functional consultants on customizing project efforts to precisely fit client manufacturing styles, across agile, lean, and just-in-time methods.

- Sole developer responsible for $20k system integration effort. Developed web services to expose sales and shipping data to external systems, along with custom hooks at AX transaction points to ensure consistency between systems.
- Proposed new AX 2012 R3 security technology to client CFO. Led the design, development, and presentation of a proof-of-concept prototype. The new solution saved $10k when compared to the alternative security options.
- Took over for senior software engineer to manage client's data integration with external Cisco phone systems. Documented and fixed over 80 service calls. Delivered operation time improvements of 50-150% on dozens of calls.

MyWebGrocer *MyWebGrocer was a leading provider of online grocery shopping*
Burlington, VT *technology for hundreds of grocery stores across the country.*

Software Developer, Intern Jun 2012 – Dec 2013

Regular contributor of code and documentation to production environments, focusing on C# and the .NET stack. Worked remotely during school year.

- Supported over 100 production websites, with zero customer complaints or issues.
- Ported 15 Classic ASP intranet tools to new international client's domain and environment. Post-completion, wrote scripts automating future transfers of these tools.

Caring for Cambodia *Caring for Cambodia is an NGO working to improve educa-*
Siem Reap, Cambodia *tion outcomes for children in Siem Reap.*
caringforcambodia.org

Senior Project for Lehigh University Sep 2013 – May 2014

Worked with team of 3 other students to design & implement a school management system for public schools in the Siem Reap province of Cambodia. Managed $6,000 IT budget. Developed project using Zend PHP, and delivered ahead of schedule and under budget.

SKILLS

- **Data storage/retrieval:** Elasticsearch, Apache Lucene, Apache Derby, PostgreSQL, MySQL, Microsoft SQL Server
- **Programming languages:** Java, Python, Typescript, Bash, Powershell, Javascript, C#, HTML5/CSS3, X++
- **Frameworks:** Spring Boot, React
- **Operating Systems:** Debian & Ubuntu, RHEL & Fedora, Windows Server, FreeBSD
- **Infrastructure:** AWS EC2 / S3 / EKS, Openstack, Kubernetes, Docker, Terraform, Ansible
- **Tools and applications:** IntelliJ IDEA, Jenkins CI, Github CI, Lens k8s IDE, Postman / HTTPie, Git

EDUCATION

Lehigh University B.A. Computer Science, May 2014
Bethlehem, PA

Coursework: Accounting, Algorithms & Data Structures, Database Design & Administration, Finance, Software Engineering, Statistics, Systems Programming